

# On the Universality of Comparability Grids for Measurement-Based Quantum Computation

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# Abstract

To be written the thesis is finalized.

# Acknowledgements

I would like to thank Professor Daniel Grier, Jackson Morris, Du (Dan) Van, Kunal Marwaha, and everyone else in the journey thus far. To be expanded when the thesis is finalized.

# Preface to the Rough Draft

Incomplete sections and comments are written in blue throughout.

One of the great difficulties writing the rough draft was figuring out where to stop. In other words, how much can I assume from the reader? To answer this question, the past honors theses of Ray Tsai, Clara Woods, and Jared Hughes in particular have been most helpful; the first for its recency and treatment of graph theory, the second for its experimental and quantum focus, and the third for its approach to 3XOR and theoretical computer science. These three, among other theses, appear to take a very introductory stance, writing for a non-expert audience familiar with only the typical mathematical fields (i.e. a senior mathematics undergraduate).

Because of this, I have prepared a (perhaps excessively) extensive preliminary section. Furthermore, I may have overextended the thesis' scope, resulting in many incomplete sections in the rough draft. I was originally planning to cut out the preliminary section entirely, or perhaps place it some extended appendix. However, no prior honors thesis (to the best of my skimming) has ever used the appendix in such a manner, nor excluded so much introduction. Thus, since it is easier to cut than to add, I have left it all in.

Furthermore, the main results of the research connected to this thesis is still a work in progress — both in the computational sense, waiting for programs to finish, and in the theoretical sense, reading up on more relevant papers and theory. Indeed, this thesis and its results are very much unfinished. Because of this, I found it challenging to create some sort of cohesive presentation of the work besides the chronological events of discoveries. Because of this, I recognize much work needs to be done to better “carve out” the original research from what is known to the field. At the same time, I feel there is merit in surveying all the developments pertaining this study, even if this dwarfs the original contributions.

# Contents

|          |                                                 |           |
|----------|-------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                             | <b>1</b>  |
| 1.1      | Informal Summary of Results . . . . .           | 2         |
| <b>2</b> | <b>Preliminaries</b>                            | <b>3</b>  |
| 2.1      | Qubits . . . . .                                | 3         |
| 2.2      | Quantum Gates . . . . .                         | 5         |
| 2.3      | Universality . . . . .                          | 6         |
| 2.4      | Measurement-Based Quantum Computation . . . . . | 7         |
| 2.5      | The ZX-Calculus . . . . .                       | 10        |
| 2.6      | The Grid as a Resource State . . . . .          | 10        |
| 2.7      | Universality Theorems . . . . .                 | 11        |
| <b>3</b> | <b>Algorithmic Approaches</b>                   | <b>13</b> |
| 3.1      | Subsets of the Comparability Grid . . . . .     | 13        |
| 3.2      | The Width-3 Gadget . . . . .                    | 18        |
| 3.3      | Universality of the Width-3 Gadget . . . . .    | 21        |
| <b>4</b> | <b>Theoretical Approaches</b>                   | <b>22</b> |
| 4.1      | LU-LC . . . . .                                 | 22        |
| 4.2      | Vertex Minors . . . . .                         | 22        |
| <b>5</b> | <b>Conclusions</b>                              | <b>24</b> |
| 5.1      | Algorithmic Results . . . . .                   | 24        |
| 5.2      | Future Work . . . . .                           | 25        |

# 1 Introduction

We start with an informal introduction to quantum computing and lay some motivations.

Fault-tolerant quantum computing has the potential, through algorithms like Shor’s factoring algorithm [17], to deliver possibly exponential speedups over classical computation. However, current Noisy Intermediate-Scale Quantum (NISQ) quantum computers have too few qubits and too much noise to deliver significant practical benefit [14].

Computation in the NISQ era is often modelled by a hybrid setup: on the quantum computer, a noisy initial state is prepared, some noisy gates are applied, and qubits are measured. A classical computer then processes these measurements, and can modify and repeat the quantum computation if necessary.

For the quantum computer, there exist a variety of computational models: The quantum Turing machine [6], the quantum circuit model [2], adiabatic quantum computation [7], measurement-based models [15], among many others. Though these models are equivalent from a complexity viewpoint (informally, any problem that is efficiently solvable in one is efficiently solvable in the others) [15, 1, 4], the underlying quantum properties used by each may differ significantly.

The measurement-based quantum computation (MBQC) model is of particular interest. This is because MBQC collapses any quantum circuit into measurements of a highly-entangled resource state that can be prepared in constant depth. This is quite shocking, as no clear classical counterpart exists. The cost of this model, however, is that measurements need to be adaptive. In other words, some post-processing must be done on-the-fly to determine what qubits should be measured next.

Since measurement is irreversible, MBQC leads to a model of one-way quantum computation. In other words, as measurement cannot increase the entanglement of a system, all entanglement required must be initially present in the resource state. Informally, entanglement is the resource consumed by this model of quantum computation. Thus, there is significant interest in determining what resource states are universal for quantum computation. That is, what resource states, under the MBQC model, can generate any possible arrangement of qubits?

The comparability grid is the graph with vertices  $V = \{(x, y) \mid x, y \in [n]\}$  and an edge from  $(x_1, y_1)$  to  $(x_2, y_2)$  if  $x_1 \geq x_2$  and  $y_1 \geq y_2$ . It is an open problem whether the family of all comparability grids is universal for MBQC. There are powerful theoretical implications for both possibilities. For example, if this family is universal, then any universal family of graph states must have unbounded rank width [cite:KunalMD]. If comparability grids are not universal, then unbounded rank width is not a sufficient condition, since the family of all comparability grids have unbounded rank width.

In this thesis, we explore the comparability grid’s potential as a resource state for universal quantum computation. In section 2, we introduce the preliminaries necessary for such an endeavour, before exploring various algorithmic and theoretical approaches in later sections. We conclude with a discussion of similar results pertaining to circle graphs.

## 1.1 Informal Summary of Results

We informally summarize the main results of this thesis. Our overall goal is to determine the universality of the comparability grids for universal MBQC. This paper will not provide a definitive result to this open problem. Instead, we will lay out a series of algorithmic and theoretical approaches to more finely characterize whether the comparability grid is a universal resource.

From the algorithmic perspective, we give a  $O(n \log n)$  algorithm to determine whether a subset of the comparability grid is “grid-like,” that is, has a few edges in addition to the regular grid. Then, the tabelau formalism for Clifford circuits and a linear-algebra-based universal enumeration algorithm is introduced to check whether this “grid-like” structure has the potential for universal quantum computation. This pipeline forms the backbone of our results.

From the theoretical perspective, we recite known results regarding local unitary and local clifford equivalence; vertex-minors; local complementation. We recall some known results on the infeasibility of an enumeration or algorithmic proof based on these mechanisms. In particular, determining vertex minors is **NP**-hard [cite:need].

We use the theoretical recitations to provide a series of conjectures on the universality of comparability grids in the conclusion section.

## 2 Preliminaries

First some notation. Let  $\mathbb{N} = \{1, 2, \dots\}$ . For  $k \in \mathbb{Z}_{\geq 0}$ , define  $[k] := \{1, \dots, k\}$ . Let  $I$  be the identity matrix of any size and of any vector space, which will be clear from context. For a graph  $G$ , let  $V(G)$  denote its vertex set and  $E(G)$  its edge set. For any vector  $v$  with  $n$  entries, let  $v_i$  be its  $i$ th entry for all  $i \in [n]$ .

### 2.1 Qubits

DEFINITION 2.1 A single **qubit**  $|\phi\rangle$  is a state

$$|\phi\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{C}^2, \quad (1)$$

where  $|\alpha|^2 + |\beta|^2 = 1$  and  $\alpha, \beta \in \mathbb{C}$ . We say  $|\phi\rangle$  is in a **superposition** between  $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$  and  $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ . Thus a single qubit is a length 2 vector in the Hilbert space  $\mathbb{C}^2$  with a  $\ell_2$  norm of 1.

DEFINITION 2.2 Define kets  $|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  and  $|1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ , both in  $\mathbb{C}^2$ . For any  $|\phi\rangle$ , let bra  $\langle\phi| := |\phi\rangle^\dagger = (|\phi\rangle^*)^T$  be its conjugate transpose. Thus the bracket denotes the inner product:

$$\langle\phi|\phi\rangle := \langle\phi| \cdot |\phi\rangle. \quad (2)$$

Also, the  $\ell_2$  norm or length  $\| |\phi\rangle \| := \sqrt{\langle\phi|\phi\rangle}$ . In general, for any  $|\phi\rangle$  and  $|\psi\rangle$ ,

$$\langle\phi|\psi\rangle := \langle\phi| \cdot |\psi\rangle = \sum_{i=1}^2 \phi_i^* \psi_i. \quad (3)$$

DEFINITION 2.3 Let  $A \in \mathbb{C}^{m \times n}$  and  $B \in \mathbb{C}^{o \times p}$ . For our purposes, it suffices to define the **tensor product**

$$A \otimes B := \begin{pmatrix} a_{1,1}B & \cdots & a_{1,n}B \\ \vdots & \ddots & \vdots \\ a_{n,1}B & \cdots & a_{n,n}B \end{pmatrix} \quad (4)$$

where  $a_{i,j}$  denotes the  $i$ th row and  $j$ th column of  $A$ . If  $x = \overline{x_1 x_2 \cdots x_d} \in \{0, 1\}^d$ ,

$$|x\rangle := \bigotimes_{i=1}^d |x_i\rangle. \quad (5)$$

We may replace  $x$  by its integer evaluation as a binary number. Also, we may write  $|\phi\rangle |\psi\rangle = |\phi\rangle \otimes |\psi\rangle$ , so adjacent kets imply  $\otimes$ . Also, let  $A^{\otimes n} := \bigotimes_{i=1}^n A$ . This gives us the definition of a  $d$ -qubit state

$$|\varphi\rangle = \sum_{x \in \{0,1\}^d} \alpha_x |x\rangle \in \mathbb{C}^{2^d}, \quad (6)$$

where  $\alpha_x \in \mathbb{C}$  and  $\sum_{x \in \{0,1\}^d} \alpha_x |x\rangle = 1$ . As the number of dimensions increases exponentially with regards to the number of qubits, bracket notation will become increasingly concise compared to traditional vector notation.

DEFINITION 2.4 Vectors  $|v_1\rangle, \dots, |v_n\rangle \in \mathbb{C}^n$  are an orthonormal basis if

1.  $\| |v_i\rangle \| = 1$  for every  $i \in [n]$  and
2.  $|v_i\rangle \perp |v_j\rangle$  (equivalently,  $\langle v_i | v_j \rangle = 0$ ) for every  $i, j \in [n]$  with  $i \neq j$ .

Henceforth, all bases are orthonormal unless explicitly said otherwise.

DEFINITION 2.5 The **Born Rule** defines how quantum measurement occurs: A  $n$ -qubit state  $|\phi\rangle$  can be measured under any basis  $|v_i\rangle \in \mathbb{C}^{2^n}$  with  $i \in [2^n]$ . When measured, the output  $i$  (interpreted as classical bits) occurs with probability  $|\langle \phi | v_i \rangle|^2$ . After seeing the outcome  $v_i$ , the state collapses to  $|v_i\rangle$ . This process is irreversible.

DEFINITION 2.6 One can also view measurement as projection. This is useful for **partial measurement**, where only some qubits in a state are measured. Let the  $n$  qubit space  $\mathbb{C}^{2^n} = S_1 \otimes S_2$  where  $S_1 \cong (\mathbb{C}^2)^k$ . Let  $\{|s_i\rangle\}_{i \in [k]}$  be a basis for  $S_1$ . Define the **projection matrix** on  $S_1$  as

$$P := \sum_{j=1}^k |s_j\rangle \langle s_j|, \quad \text{such that} \quad P = P^\dagger, \quad P^2 = P. \quad (7)$$

Define the **partial measurement** on  $S_1$  by  $M = P \otimes I \in \mathbb{C}^{2^n \times 2^n}$ . For a state  $|\phi\rangle \in \mathbb{C}^{2^n}$ ,

$$\Pr(\text{Measuring } i \text{ from } |\phi\rangle) = \langle \phi | M | \phi \rangle = \|M |\phi\rangle\|^2. \quad (8)$$

On measuring outcome  $i$ , the post-measurement state becomes

$$|\phi_i\rangle = \frac{M |\phi\rangle}{\sqrt{\langle \phi | M | \phi \rangle}}. \quad (9)$$

In other words,  $|\phi_i\rangle$  is the state that minimizes  $\| |\phi\rangle - |\phi_j\rangle \|$  over all  $|\phi_j\rangle \in S_1$ . Partial measurements technically give rise to mixed states, which cannot be represented as a single ket, but we defer its discussion to a later section.

EXAMPLE 2.7 Consider the states

$$|+\rangle := \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle, \quad |-\rangle := \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle. \quad (10)$$

Measuring either in the **standard basis**  $\{|0\rangle, |1\rangle\}$  gives  $|0\rangle$  with 0.5 probability and  $|1\rangle$  with 0.5 probability since

$$|\langle + | 0 \rangle|^2 = \left| \frac{1}{\sqrt{2}} \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 \right|^2 = \frac{1}{2}$$

and similarly for  $|1\rangle$ . The squaring makes  $|-\rangle$  proceed similarly as well.

DEFINITION 2.8 A two qubit state  $|\phi\rangle$  is **entangled** if it cannot be written as the tensor product of single qubits. That is,  $\nexists |\psi\rangle, |\varphi\rangle \in \mathbb{C}^2$  such that  $|\phi\rangle = |\psi\rangle \otimes |\varphi\rangle$ . If a state is not entangled, it is called **separable**.

## 2.2 Quantum Gates

DEFINITION 2.9 A single qubit quantum gate  $U \in \mathbb{C}^{2 \times 2}$  satisfies one of:

1.  $U^\dagger U = UU^\dagger = I$ ;
2. equivalently,  $U$  is a unitary matrix;
3. equivalently,  $U = |\phi_0\rangle\langle 0| + |\phi_1\rangle\langle 1|$ , where  $\{|\phi_0\rangle, |\phi_1\rangle\}$  form an orthonormal basis.

Common single qubit gates are given in Figure 1.

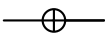
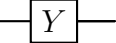
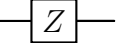
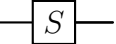
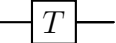
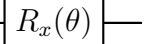
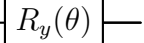
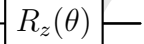
| Name          | Circuit                                                                             | Matrix                                                                                                 | Bra-Ket                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X$           |    | $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$                                                         | $ 1\rangle\langle 0  +  0\rangle\langle 1 $                                                                                                           |
| $Y$           |    | $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$                                                        | $i 1\rangle\langle 0  - i 0\rangle\langle 1 $                                                                                                         |
| $Z$           |    | $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$                                                        | $ 0\rangle\langle 0  -  1\rangle\langle 1 $                                                                                                           |
| $H$           |   | $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$                                     | $ +\rangle\langle 0  +  -\rangle\langle 1 $                                                                                                           |
| $S$           |  | $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$                                                         | $ 0\rangle\langle 0  + i 1\rangle\langle 1 $                                                                                                          |
| $T$           |  | $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$                                                | $ 0\rangle\langle 0  + e^{i\pi/4} 1\rangle\langle 1 $                                                                                                 |
| $R_x(\theta)$ |  | $\begin{pmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$ | $\cos(\theta/2) 0\rangle\langle 0  - i\sin(\theta/2) 1\rangle\langle 0 $<br>$-i\sin(\theta/2) 0\rangle\langle 1  + \cos(\theta/2) 1\rangle\langle 1 $ |
| $R_y(\theta)$ |  | $\begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$    | $\cos(\theta/2) 0\rangle\langle 0  + \sin(\theta/2) 1\rangle\langle 0 $<br>$-\sin(\theta/2) 0\rangle\langle 1  + \cos(\theta/2) 1\rangle\langle 1 $   |
| $R_z(\theta)$ |  | $\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$                                | $e^{-i\theta/2} 0\rangle\langle 0  + e^{i\theta/2} 1\rangle\langle 1 $                                                                                |

Figure 1: Common single qubit quantum gates

FACT 2.10 For matrices  $A, B, C, D$ , scalar  $\lambda$ , and vectors  $|\phi\rangle, |\psi\rangle$ ,  $\otimes$  satisfies:

1. Bi-linearity:  $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ , so  $(A \otimes B)(|\phi\rangle \otimes |\psi\rangle) = (A|\phi\rangle) \otimes (B|\psi\rangle)$ .
2. Scalars commute:  $\lambda(A \otimes B) = (\lambda A) \otimes B = A \otimes (\lambda B)$ .
3. Associativity:  $(A \otimes B) \otimes C = A \otimes (B \otimes C)$ .
4. Unitary-preserving: If  $A, B$  are unitary matrices,  $A \otimes B$  is unitary.

DEFINITION 2.11 The general 1-qubit unitary is given by

$$U(\theta, \iota, \kappa) := \begin{pmatrix} \cos(\frac{\theta}{2}) & -e^{i\kappa} \sin(\frac{\theta}{2}) \\ e^{i\iota} \sin(\frac{\theta}{2}) & e^{i(\iota+\kappa)} \cos(\frac{\theta}{2}) \end{pmatrix}. \quad (11)$$

FACT 2.12 Any 1-qubit gate can be written as  $U(\theta, \iota, \kappa)$  for some  $\theta, \iota, \kappa \in [0, 2\pi]$  [cite:need].

DEFINITION 2.13 An  **$n$ -qubit gate** is a unitary matrix in  $\mathbb{C}^{2^n \times 2^n}$ . If gates  $U_1, \dots, U_k$  satisfy  $U = \bigotimes_{i=1}^k U_i \in \mathbb{C}^{2^n}$ , by Fact 2.10.4  $U$  is an  $n$ -qubit gate. Special named multi-qubit gates are given in Figure 2.

| Name             | Circuit | Bra-Ket                                                                                                                   |
|------------------|---------|---------------------------------------------------------------------------------------------------------------------------|
| CNOT             |         | $ 0\rangle\langle 0  \otimes I +  1\rangle\langle 1  \otimes X$                                                           |
| CCNOT or Toffoli |         | $ 11\rangle\langle 11  \otimes X + \sum_{\substack{(a,b) \neq (1,1) \\ a,b \in \{0,1\}}}  ab\rangle\langle ab  \otimes I$ |
| SWAP             |         | $\sum_{i,j \in \{0,1\}}  ij\rangle\langle ji $                                                                            |
| C-Z              |         | $ 0\rangle\langle 0  \otimes I +  1\rangle\langle 1  \otimes Z$                                                           |

Figure 2: Common multi-qubit quantum gates

DEFINITION 2.14 Many of the gates in Figure 2 are **controlled** gates. Informally, controlled gates, such as  $\text{---}\oplus\text{---}$  only trigger if all of the controls  $\text{---}\bullet\text{---}$  connected to it are  $|1\rangle$ . Formally, for any  $n$ -qubit gate  $U$ , the single controlled gate, denoted C- $U$ , is defined

$$\text{C-}U := |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U. \quad (12)$$

This definition generalizes for multiple controls. Thus CNOT is just C- $X$  and Toffoli is CC- $X$ . We may suppress the - if the meaning is clear. C- $Z$  is special as it only triggers if both qubits are  $|1\rangle$ . Note that  $I = |0\rangle\langle 0| + |1\rangle\langle 1|$  so  $Z = I - 2|1\rangle\langle 1|$ . Then

$$\text{C-Z} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes (I - 2|1\rangle\langle 1|) = I \otimes I - 2|11\rangle\langle 11|.$$

Thus, it can be interpreted that both qubits of the C- $Z$  are controls.

## 2.3 Universality

DEFINITION 2.15 For every  $n \in \mathbb{N}$ , Let  $\text{SU}(n)$  denote the special unitary group under matrix multiplication consisting of  $n \times n$  unitary matrices with determinant 1. It is a normal subgroup of the unitary group  $U(n)$ , consisting of  $n \times n$  unitary matrices.

**DEFINITION 2.16** A set of gates  $\mathcal{S}$  is (efficiently) **universal** if any  $U \in \text{SU}(2^n)$ , for any  $n \in \mathbb{N}$ , that can be generated (in  $\text{poly}(n)$  gates) by an arbitrary gate set  $\mathcal{S}'$ , can also be generated by the gate set  $\mathcal{S}$  (in  $\text{poly}(n)$  gates) [12]. We say a set of gates  $\mathcal{S}$  is  $d$ -qubit universal if the above holds for  $U \in \text{SU}(2^d)$ .

**REMARK 2.17** In this thesis we will only focus on universality, not efficient universality. Thus, it may take exponential or more gates to simulate certain unitaries  $U$ . Furthermore, in practice, the application of  $U$  up to some measure and error  $\epsilon$  suffices. This is called **approximate universality**. This thesis will not focus on approximate universality either.

**REMARK 2.18** Definition 2.16 is only concerned with implementing arbitrary unitaries, not necessarily computation. That is, it does not specify whether the inputs and outputs are classical or quantum in nature. This matters because  $|\phi\rangle = |0\rangle$  and  $|\psi\rangle = -|0\rangle$  have the same measurement distribution and classical output, but  $|\phi\rangle \neq |\psi\rangle$ . This difference is called **global phase**, i.e. a scalar  $e^{i\theta}$  multiplied to an entire state.

**DEFINITION 2.19** We use the following types of **universal computation** from [12]:

- **CC-universal**: For every  $U \in \text{SU}(n)$ , the model can reproduce the distribution of local measurements performed on  $U\{|0^n\rangle\}$ .
- **QC-universal**: Given an arbitrary  $n$ -qubit state  $|\phi\rangle$ , for every  $U \in \text{SU}(2^n)$ , the model can reproduce the distribution of local measurements performed on  $U|\phi\rangle$ .
- **CQ-universal**: For every  $U \in \text{SU}(2^n)$ , the model can produce the quantum state  $|\phi\rangle = U|0^n\rangle$ .
- **QQ-universal**: Given an arbitrary  $n$ -qubit state  $|\phi\rangle$ , for every  $U \in \text{SU}(2^n)$ , the model can produce the quantum state  $U|\phi\rangle$ .

**FACT 2.20** Any 1-qubit universal gate set and a 2 qubit entangling gate is universal [cite:need].

**FACT 2.21** Any universal gate set must include at least one multi-qubit entangling gate, as it is impossible to entangle using only single-qubit gates.

**EXAMPLE 2.22** The **Clifford gates** consist of Hadamard  $H$ , Phase  $S$ , and CNOT. The Gottesman-Knill theorem [10] shows this gate set is not universal and is classically efficiently simulable.

## 2.4 Measurement-Based Quantum Computation

**DEFINITION 2.23** Measurement-based quantum computation (MBQC) roughly proceeds as follows:

1. A classical input specifies the data and program.

2. A resource state  $|\phi_k\rangle$  is chosen from some set of states  $\Phi = \{|\phi_1\rangle, |\phi_2\rangle, \dots\}$ .
3. Local operations (single qubit unitaries), local measurements, and classical communication are used to transform this resource state.
4. By the end, we have a state  $|\psi\rangle \otimes |\text{junk}\rangle$ . This  $|\psi\rangle$  or its measurement is the desired result.

The rest of this section formalizes these notions.

**DEFINITION 2.24** A resource state is a quantum state  $|\phi\rangle$  that can be used as the initial state in MBQC. A **graph state**  $|G\rangle$  is a particular type of resource state associated with an undirected graph  $G = (V, E)$  such that:

- For every  $v \in V(G)$ , create a new qubit in the  $|+\rangle$  state using  $H$ .
- For every  $e = (u, v) \in E(G)$ , apply C-Z on qubits associated with  $u$  and  $v$ .

**FACT 2.25** There is a unique construction for every graph state, up to reordering of C-Z. This is because C-Zs commute with  $I$  and C-Z.

*Proof.* The C-Z gate is a diagonal matrix, and so is  $I$ . The tensor product of diagonal matrices is a diagonal matrix. Furthermore, diagonal matrices commute. Thus the order of adding edges, or equivalently applying C-Zs, does not matter.  $\square$

**FACT 2.26** Let  $G$  be a graph with maximum degree  $\Delta$ . Then  $|G\rangle$  can be constructed in depth  $\Delta + 1$ .

*Proof.* Vizing's theorem states that any graph with maximum degree  $\Delta$  admits an edge coloring with  $\Delta + 1$  colors such that no two edges that share a vertex have the same color. All edges of the same color can be implemented in the graph state in the same layer of C-Zs. Thus, at most  $\Delta + 1$  layers.  $\square$

**FACT 2.27** Define the Hadamard gadget as the following circuit:

$$\begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \boxed{H} \\ \text{---} \end{array} \begin{array}{c} \boxed{\text{Measurement}} \\ \text{---} \end{array} \quad (13)$$

Commuting a Hadamard through the C-Z gives a CNOT:

$$\begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \boxed{H} \\ \text{---} \end{array} \begin{array}{c} \boxed{\text{Measurement}} \\ \text{---} \end{array} = \begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \boxed{H} \\ \text{---} \end{array} \begin{array}{c} \oplus \\ \bullet \end{array} \begin{array}{c} \boxed{\text{Measurement}} \\ \text{---} \end{array} = \begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \oplus \\ \bullet \end{array} \begin{array}{c} \boxed{\text{Measurement}} \\ \text{---} \end{array} \quad (14)$$

Two CNOTs cancel each other out and a CNOT has no effect on  $|+\rangle$ . Then

$$\begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \oplus \\ \bullet \end{array} \begin{array}{c} \boxed{\text{Measurement}} \\ \text{---} \end{array} = \begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \oplus \\ \bullet \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \boxed{\text{Measurement}} \\ \text{---} \end{array} \quad (15)$$

The first three CNOTs are equivalent to a swap gate. Thus

$$\begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \\ \bullet \\ \oplus \\ \bullet \\ \oplus \\ \bullet \\ \oplus \end{array} \begin{array}{c} \oplus \\ \bullet \\ \oplus \\ \bullet \\ \oplus \\ \bullet \\ \oplus \\ \bullet \end{array} \begin{array}{c} \oplus \\ \bullet \\ \oplus \\ \bullet \\ \oplus \\ \bullet \\ \oplus \\ \bullet \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \\ \\ \\ \\ \\ \end{array} = \begin{array}{c} |+\rangle \\ H|\phi\rangle \end{array} \begin{array}{c} \oplus \\ \bullet \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array} \quad (16)$$

The final state before measurement is

$$\text{CNOT}(|+\rangle \otimes H|\phi\rangle) = \frac{1}{\sqrt{2}} (|0\rangle \otimes H|\phi\rangle + |1\rangle \otimes XH|\phi\rangle). \quad (17)$$

In other words, measuring in the  $X$  basis ( $\{|0\rangle, |1\rangle\}$ ) applies an  $X$  to  $|\phi\rangle$  if we measure a  $|1\rangle$ . This happens with only 0.5 probability, hence the need for **adaptive measurements**. That is, measurements whose classical outcome can dynamically impact how other qubits are measured.

**FACT 2.28**  $R_z(\theta)ZR_z(\theta)^\dagger = Z$ . That is,  $R_z(\theta)$  commutes with C-Z for all  $\theta$ . Thus,

$$\begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{R_z(\theta)} \\ \\ \end{array} \begin{array}{c} \boxed{H} \\ \\ \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array} = \begin{array}{c} |+\rangle \\ HR_z\theta|\phi\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array}.$$

**FACT 2.29** [Euler decomposition] Every single-qubit unitary can be expressed (up to global phase) as the product  $R_z(\theta_3)R_x(\theta_2)R_z(\theta_1)$ .

**FACT 2.30** For all  $\theta \in [0, 2\pi]$ ,

$$R_x(\theta) = HR_z(\theta)H. \quad (18)$$

Thus by Fact 2.29 every single-qubit unitary can be applied to any  $|\phi\rangle$  given we are making the right measurements.

**THEOREM 2.31** [MBQC on the line is 1-qubit CQ-universal] Suppose you wish to apply gates  $U_1, \dots, U_n$  to  $|+\rangle$ . There is a sequence of single-qubit measurements you can do to the line graph state of  $|V| = \Theta(n)$  such that, assuming measurements are all 0, gives  $U_1, \dots, U_n |+\rangle$  on the last qubit.

The previous statement is really more of a statement on the computational power (in terms of universality) of the line graph as opposed to MBQC itself. For example, the graph with a single vertex is clearly not universal with the same MBQC gadgets. We formalize this notion below:

**THEOREM 2.32** [[12] Observation 2: CQ-universal implies QQ-universal]. A resource is CQ-universal if, using the MBQC protocol (namely, local operations and measurements), we obtain CQ-universality. Every CQ-universal resource can be made QQ-universal by allowing quantum inputs to be coupled into the resource state using the same coupling method as before, i.e. C-Z gates.

## 2.5 The ZX-Calculus

This section is meant to introduce the ZX-calculus, a graphical representation of quantum circuits that makes MBQC in particular far easier to analyze. It would be most helpful for proofs in later sections as well. That said, it requires significant diagramming which I have unfortunately not done (and am still quite slow at). I am considering referring readers to the excellent “ZX-calculus for the working computer scientist” by J. van de Wetering [20]. If the preliminary section survives the rough draft, then I will give a more formal introduction of the ZX-calculus here with a focus on MBQC and its use in finding graph identities.

In particular, I would like to introduce the following ZX-rules: (the following figure is a screenshot from [8])

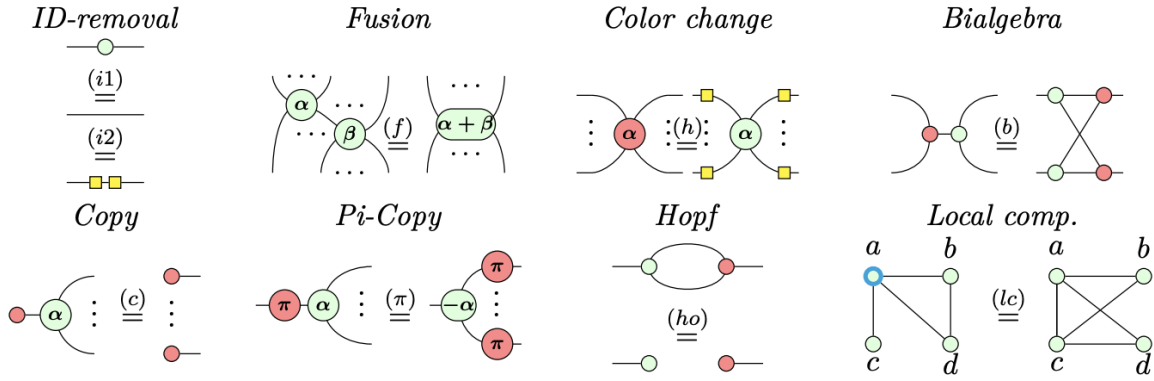


Figure 5: The basic rewriting rules of ZX-calculus.

Figure 3: The rules of ZX-calculus, taken from [8]

## 2.6 The Grid as a Resource State

DEFINITION 2.33 A  $m \times n$  grid graph  $G = (V, E)$  has

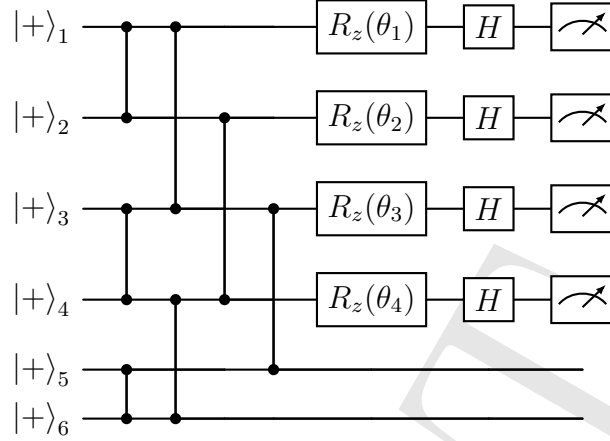
$$V = \{(i, j) \mid i \in [m], j \in [n]\}, \quad (19)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, |i_2 - i_1| + |j_2 - j_1| = 1\} \quad (20)$$

In other words,  $m$  rows and  $n$  columns with edges between nodes of taxicab distance 1. Denote such graphs as  $\text{Grid}_{m \times n}$ . Let  $\text{Grid}$  denote the family of all such graphs.

THEOREM 2.34  $\text{Grid}$  is 2-qubit QQ-universal. That is, given a set of 2-qubit gates  $U_1, \dots, U_k$  applied to the  $|++\rangle$  state, there is a sequence of single qubit measurements you can make on some graph state in  $\text{Grid}$ , assuming all measurement results are 0, resulting in  $U_1 \cdots U_k |++\rangle$  on the last

*Proof.* By Theorem 2.32, it suffices to show CQ-universality. Consider the following construction:



The above construction can be repeated to apply  $R_z$  as many times as necessary. The analysis is similar to Theorem 2.31. That is, it suffices to show that every 2-qubit gate can be decomposed into a sequence of

$$(H \otimes H) (R_z(\theta_1) \otimes R_z(\theta_2)) \text{ C-Z.} \quad (21)$$

The proof of this is essentially brute force [12]. [This thesis' ultimate goal is to recreate this proof, so it should be reproduced here.](#)  $\square$

**THEOREM 2.35** Grid is QQ-universal.

*Proof.* [To be done. I had something written here, but I decided to rewrite it using some lemmas more easily proved in the ZX-calculus.](#)  $\square$

**DEFINITION 2.36** Let LO denote local operations, i.e. any single qubit unitary. Let CC denote classical communication. Thus MBQC requires LOCC, or both LO through Fact 2.30 and CC due to adaptive measurement.

**THEOREM 2.37** [From [12] Observation 5] All efficient universal graph state resources can efficiently be obtained from each other by LOCC.

**REMARK 2.38** The previous theorem informally says that the entanglement structure of a resource state is what matters, not its phase. This is because LOCC are precisely the operations that do not add entanglement [cite:need].

## 2.7 Universality Theorems

**DEFINITION 2.39** A family of states  $\Phi$  is a **universal resource** for MBQC if for all  $n$ -qubit state  $|\psi\rangle$ ,  $\exists |\phi\rangle \in \Phi$  with  $m \geq n$  qubits, such that the transformation  $|\phi\rangle \mapsto |\psi\rangle |+\rangle^{m-n}$  is possible with probability 1 with LOCC. We denote this by  $|\phi\rangle \geq_{\text{LOCC}} |\psi\rangle$ .

THEOREM 2.40 [From [19]] Let  $\Psi$  be a family of states and  $|\phi\rangle, |\phi'\rangle$  arbitrary states with the same number of qubits. Let  $E : |\phi\rangle \rightarrow \mathbb{R}$  be a function that satisfies the following properties:

1.  $E(|\phi\rangle) \geq E(|\phi'\rangle)$  whenever  $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$ ;
2.  $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$ .

Then  $\Psi$  cannot be a universal resource.

*Proof.* The first condition states that LOCC cannot increase this measure, while the second says that the supremum value of  $E$ , which can be  $\infty$ , is not reached among all  $|\psi\rangle \in \Psi$ . If both were true, then (CQ)-universality would be violated.  $\square$

REMARK 2.41  $E$  as a measure is so general there are many candidates. However, satisfying the converse of the second condition is not sufficient to guarantee universality. For example, the toric code states have unbounded entanglement width but are not universal [3].

There is more to be said here, but I'm not sure how much of this additional material is necessary, so I've left this out for now.

### 3 Algorithmic Approaches

DEFINITION 3.1 The  $m \times n$  **comparability grid** is a graph  $G = (V, E)$  with

$$V = \{(i, j) \mid i \in [m], j \in [n]\} \quad (22)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, i_2 \geq i_1, j_2 \geq j_1\} \quad (23)$$

In other words,  $m$  rows and  $n$  columns with edges defined by the partial order  $\leq$ . Denote such graphs as  $\text{CGrid}_{m \times n}$ . Let  $\text{CGrid}$  denote the family of all such graphs. By abuse of notation, let it also denote the resource state of all graph states corresponding to comparability grids.

EXAMPLE 3.2 Figure 3 shows the  $3 \times 3$  comparability grid:

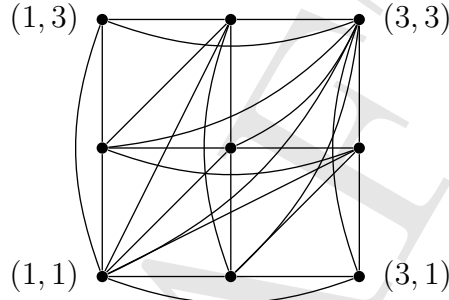


Figure 4:  $\text{CGrid}_{3 \times 3}$ .

We dedicate the rest of this section to algorithmic approaches in determining the CQ-universality of the  $\text{CGrid}$  resource.

#### 3.1 Subsets of the Comparability Grid

One idea it to ask how close “vertex deletion” can get us to universal quantum computation. Namely, can we delete vertices to get a  $\text{Grid}$ ? If not, how close can we get?

DEFINITION 3.3 Define a subset of a  $\text{CGrid}$  state as the graph corresponding to a state in  $\text{CGrid}$  with some vertices measured in the  $Z$  basis. Graph theoretically, a subset of a  $\text{CGrid}$  is a  $\text{CGrid}$  with some vertices removed (alongside any edges connected to said removed vertices).

FACT 3.4 Any subset  $H$  with  $n$  vertices of any  $\text{CGrid}$  can be uniquely identified as a permutation of  $[n]$ , denoted  $\pi \in S_n$ . Denote this by  $\pi(H)$ . Similarly, let  $H(\pi)$  denote the subgraph corresponding to  $\pi$ .

*Proof.* This proof only works for finite subsets, but there is a need to define it for infinite-sized vertex subsets. Consider an arbitrary subset  $H = (V, E)$  with  $|V| = n$  of a fixed  $\text{CGrid}$   $G$ . We claim this subgraph is isomorphic to the graph  $H' = (V', E')$ , where

$$1. V' = \{(x + y/n, y + x/n) \mid (x, y) \in V\}$$

$$2. E' = \{ \{ (x_1 + y_1/n, y_1 + x_1/n), (x_2 + y_2/n, y_2 + x_2/n) \} \mid \{ (x_1, y_1), (x_2, y_2) \} \in E \}.$$

That is,  $H'$  is the graph whose  $x'$  and  $y'$  values have been shifted rightwards and upwards, respectively. Since  $|V| = n$ , there is a clear bijection from  $V$  to  $V'$ . The bijection between edges follows as  $|E| = |E'|$ .

We claim the data of  $H'$  can be completely expressed as follows: Sort the vertices in  $V'$  by their  $x'$ -coordinate. No two distinct vertices can share an  $x'$  coordinate: We proceed by contradiction. If  $x_1 = x_2$

$$x_1 + \frac{y_1}{n} = x_2 + \frac{y_2}{n} \implies y_1 = y_2 \quad (24)$$

but then they are not distinct vertices. If  $x_1 \neq x_2$ , without loss of generality let  $x_1 > x_2$  so  $x_1 = x_2 + c$ ,  $c \in \mathbb{N}$ . Then

$$x_1 + \frac{y_1}{n} = x_2 + \frac{y_2}{n} \implies c = \frac{y_1}{n} - \frac{y_2}{n} \implies c = \frac{1}{n}(y_1 - y_2) \quad (25)$$

Since  $y_1, y_2 \leq n$ , the RHS  $\leq 1$ . Contradiction. No two distinct vertices share a  $y'$  coordinate either, by a similar proof. Re-label each vertex by their order, from least to greatest, in both  $x'$  and  $y'$ . Edges still form by the partial ordering rule in Equation 23. Reading the  $x'$  coordinate as the index of a  $n$ -tuple and the  $y'$  coordinate as the values, we obtain a permutation of  $n$  as each  $x'$ ,  $y'$  is distinct.  $\square$

**FACT 3.5** Two subsets  $H, H'$  of a CGrid may have  $\pi(H) \neq \pi(H')$  yet  $H = H'$  up to vertex relabeling.

*Proof.* Take subgraphs with vertices  $V = \{(1, 1), (1, 2)\}$  and  $V' = \{(1, 1), (2, 1)\}$  and their induced edges. The first has permutation  $(1, 2)$  and the second  $(2, 1)$  but the graphs are clearly isomorphic.  $\square$

**FACT 3.6** Let  $\pi \in S_n$  and define its reverse-complement

$$\pi^{\text{rc}}(i) := (n + 1) - \pi(n + 1 - i). \quad (26)$$

Then the induced CGrid subgraphs satisfy  $H(\pi) \cong H(\pi^{\text{rc}})$ .

*Proof.* Write vertices of  $H(\pi)$  as  $v_i = (i, \pi(i))$ . Define a bijection

$$f : v_i \mapsto v'_{n+1-i} := (n + 1 - i, n + 1 - \pi(i)). \quad (27)$$

The image vertex set is exactly  $\{(k, \pi^{\text{rc}}(k)) \mid k \in [n]\}$ , so  $f$  is a bijection from  $V(H(\pi))$  to  $V(H(\pi^{\text{rc}}))$ . An edge in a comparability grid exists between  $v_i$  and  $v_j$  iff

$$(i - j)(\pi(i) - \pi(j)) > 0, \quad (28)$$

equivalently  $(i < j \ \& \ \pi(i) < \pi(j))$  or  $(i > j \ \& \ \pi(i) > \pi(j))$ . But under  $f$  we have

$$(n + 1 - i) - (n + 1 - j) = -(i - j), \quad (29)$$

$$(n + 1 - \pi(i)) - (n + 1 - \pi(j)) = -(\pi(i) - \pi(j)), \quad (30)$$

so the product  $(i - j)(\pi(i) - \pi(j))$  is preserved. Hence  $\{v_i, v_j\}$  is an edge iff  $\{f(v_i), f(v_j)\}$  is an edge. Therefore  $f$  is a graph isomorphism.  $\square$

**THEOREM 3.7** Let  $\pi(H)$  denote a permutation of  $n$  capturing the data of a finite subset  $H = (V, E) \leq G$  with  $|V| = n$ , for some  $G \in \text{CGrid}$ . The degree sequence of  $\pi(H)$  can be constructed algorithmically in  $O(n \log n)$  time and  $O(n)$  space.

*Proof.* For any  $\pi \in S_n$ , let an **inversion** be a pair  $(i, j)$ ,  $i \neq j \in [n]$ , such that

$$i < j \text{ and } \pi[i] > \pi[j]. \quad (31)$$

$(i, j)$  is a **non-inversion** if

$$i < j \text{ and } \pi[i] < \pi[j]. \quad (32)$$

Since  $i \neq j$ , every  $(i, j)$  is either an inversion or non-inversion. Thus

$$\#\text{non-inversions} = \binom{n}{2} - \#\text{inversions}. \quad (33)$$

At position  $j \in [n]$  of the permutation,

$$|\{i < j \mid \pi[i] < \pi[j]\}| = \text{prefixSum}(\pi[j] - 1). \quad (34)$$

Here, prefix sum refers to the number of indices with index  $\leq j$  and value  $\leq \pi[j]$ . The  $-1$  makes  $\leq$  become  $<$ . In other words, for each  $j$ , we count the number of indices  $i < j$  such that  $\pi[i] < \pi[j]$ .

It is well-known that a prefix sum can be computed in  $\Theta(\log n)$  time using a binary index tree or Fenwick tree. Since there are  $n$  iterations, the total time is  $\Theta(n \log n)$ . The Fenwick tree is the only allocation, and only needs  $\Theta(n)$  more space.

Run the above procedure on the reverse of  $\pi(H)$ . This counts, for each  $j$ , the number of indices  $i < j$  such that  $\pi[i] > \pi[j]$ . Adding the two values for each index gives the degree of the corresponding vertex. Denote this algorithm as **double inversion**.  $\square$

We use double-inversion rather than directly constructing the edges due to the following fact:

**FACT 3.8** In a subgraph  $H$  with  $n$  vertices, there are at most  $O(n^2)$  edges. Thus, enumerating the edges of some  $\pi(H)$  cannot be better than  $O(n^2)$  in the worst general case.

**THEOREM 3.9** It is sufficient to explore subgraphs with at most  $O(n)$  edges. Thus, enumerating the edges of some  $\pi(H)$  cannot be worse than  $O(n)$  in the worst general case.

*Proof.* **This needs some formalization.** The idea is to use early termination. Suppose we identify, even in the first pass of Theorem 3.7, that the current graph has “too many” edges or “too many” vertices with “high degree” to be the construction with optimal minimum number of edges compared to the **Grid**. Since every iteration of the algorithm through the vertices in both passes can only add 0 or more edges, we can early terminate the entire branch. We call these early terminations **termination strategies**.  $\square$

**THEOREM 3.10** When enumerating every possible subgraph  $H = (V, E)$  of some CGrid with  $|V| = n$ , it is more than sufficient to enumerate every possible permutation of  $[n]$ .

*Proof.* Clearly, there is a map  $f : \{\forall H\} \rightarrow S_n$ . If there is a subset that does not have a representation in  $S_n$ , this would violate Fact 3.4. This is more than sufficient as Fact 3.5 shows the permutation representation is not unique.  $\square$

Before we enumerate the termination strategies we will use, it may be helpful to define some notation.

**DEFINITION 3.11** Given any Grid, say a vertex is a corner vertex if it has degree 2 in the grid. Say it is an edge if it has degree 3, and inner if it has degree 4. In the context of a subgraph  $H$  of some CGrid, say a vertex in  $H$  is a **potential corner** if it has degree at least 2. Similarly for **potential edge** and **potential inner**.

Furthermore, say a vertex in  $H$  is **invalid** if it has degree less than 2. We say invalid as we wish to find the number of maximum additional edges, not minimum number of missing edges. In other words, if  $H$  has an invalid vertex, no Grid  $G$  with  $|V(H)| = |V(G)|$  is a subgraph of  $H$ .

**FACT 3.12** To enumerate all possible permutations of  $[n]$ , it suffices to consider all possible choices for the first entry, then all possible remaining choices for the second entry for each chosen first entry, and henceforth. This task is massively parallelizable by seeding  $k \leq n$  first entries.

*This needs to be made more formal.* Thus we can say we **terminate** a set of permutations in  $S_n$  if the processing of  $j \leq n$  entries in either pass of Theorem 3.7 triggers a termination strategy.

**DEFINITION 3.13** *This needs to be edited for brevity.* We now enumerate the various termination strategies. Throughout, let  $H = (V, E)$  be the subgraph of some CGrid, and let  $\text{Grid}_{k \times n}$  be the grid we are comparing  $H$  with. Let first and second pass refer to the two passes of Theorem 3.7.

Suppose we set a bound  $b$  on the maximum number of edges.

1. If in the second pass, a vertex is found invalid, terminate. No vertex can be found invalid in the first pass.
2. If the sum of degrees, divided by 2 due to the Handshaking lemma, is larger than  $b$ , terminate.
3. Suppose a vertex has degree  $> b + 4$ . Then this subgraph invalidates  $b$ . *This may be equivalent to the previous strategy.*
4. The distribution of degrees should be approximately constant. In other words, the additional edges  $b$  should be spread around the graph. This can be computed by keeping a counter of the number of degree  $1, \dots, b + 4$  vertices found in the second pass.

**THEOREM 3.14** All the termination strategies above can be done in  $\Theta(1)$  time and  $\Theta(1)$  space.

*Proof.* Most strategies can be done in such bounds by keeping a running sum, running a constant number of if statements, and other  $\Theta(1)$  time and space operations.  $\square$

Recall that our goal is to identify whether vertex deletion alone can get us to a subgraph isomorphic to some **Grid** graph. This is surprisingly true for  $2 \times n$  grids for all  $n \in \mathbb{N}$ .

**THEOREM 3.15** The  $2 \times n$  **Grid** is obtainable from a sufficiently large **CGrid** through vertex deletion alone.

*Proof.* Fix  $G = \text{CGrid}_{n \times n}$ . Define the  $i$ th **anti-diagonal** as

$$A_n := \{(i, j) \mid i + j = n + 1\} \quad (35)$$

For example,  $A_1 = \{(1, 1)\}$  and  $A_2 = \{(1, 2), (2, 1)\}$ . Delete all vertices not in some  $A_j$  or  $A_{j+1} \setminus \{(j, 1), (1, j)\}$  for  $j \in [n - 1]$ . The remaining vertices and induced edges form a  $2 \times n$  grid:

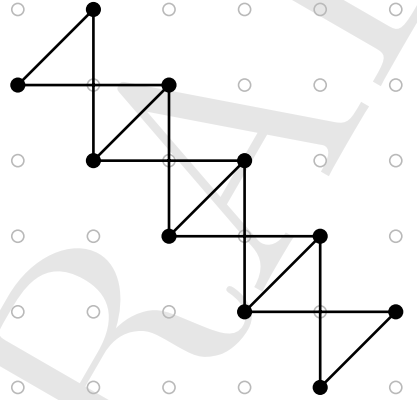


Figure 5: The  $2 \times 5$  **Grid** subset in a  $\text{CGrid}_{6 \times 6}$  using only vertex deletion.

To see that this is a grid, “twist” the above figure.  $\square$

However, the above pattern does not generalize for larger widths.

**FACT 3.16** It is not possible to get a  $\text{Grid}_{3 \times n}$  subgraph from any size of **CGrid** using only vertex deletion.

*Proof.* Brute force using the algorithm described in Theorem 3.7.  $\square$

That said, we can get quite close to a grid:

**DEFINITION 3.17** Consider the following construction:

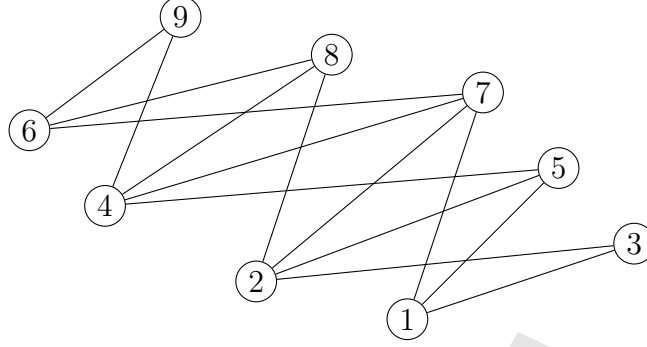


Figure 6: A subset close to a  $\text{Grid}_{3 \times 3}$  from some large  $\text{CGrid}$  using only vertex deletion.

This forms the following subgrid:

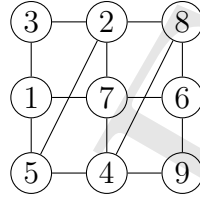


Figure 7: Tileable  $\text{Grid}_{3 \times n}$  subgraph of  $\text{CGrid}$  with one additional edge every column.

This construction is tileable. That is, it generalizes to  $\text{Grid}_{3 \times n}$  provided a sufficiently large  $\text{CGrid}$ .

*Proof.* I had an extremely complicated proof. The crux of the issue is that every time a new column is added, we need to ensure that it can "fit" in between (in either the x or y dimension) some previous existing points. Since our points need to be in  $\mathbb{N} \times \mathbb{N}$ , the simplest solution is to pre-allocate some gap between the nodes. However, this becomes extremely cumbersome very quickly.

An easier proof is to work graphically. That said, it would be great if the proof was in the ZX-representation of the  $\text{CGrid}$ , as this may highlight some important identities and tricks usable for even higher widths.  $\square$

### 3.2 The Width-3 Gadget

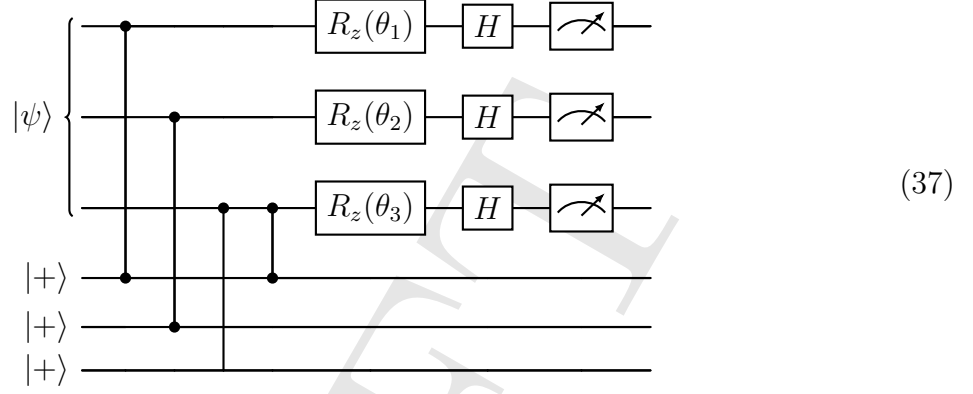
The natural question to ask next is whether the additional edge poses a problem for universality. First we begin with a useful fact.

FACT 3.18 The following circuit identity holds:

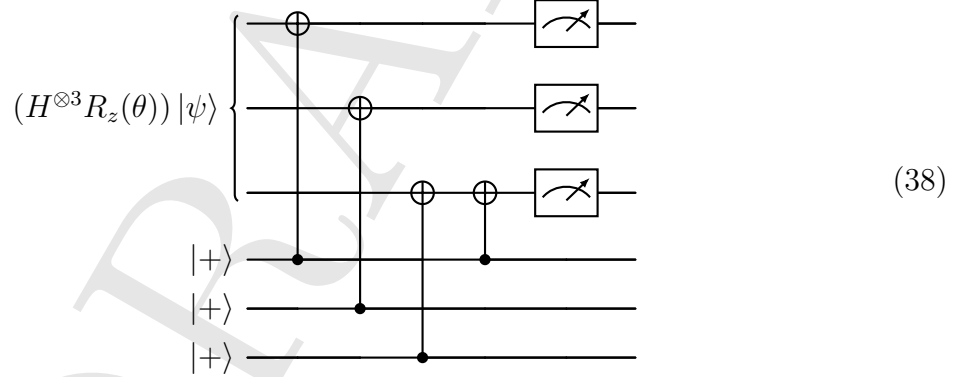
$$\begin{array}{c}
 q_0 \text{ --- } \bullet \text{ --- } \oplus \\
 | \quad | \\
 q_1 \text{ --- } \bullet \\
 | \\
 q_2 \text{ --- } \oplus
 \end{array}
 =
 \begin{array}{c}
 q_0 \text{ --- } \oplus \text{ --- } \bullet \\
 | \quad | \\
 q_1 \text{ --- } \bullet \text{ --- } \bullet \\
 | \quad | \\
 q_2 \text{ --- } \oplus \text{ --- } \oplus
 \end{array}
 \tag{36}$$

*Proof.* By enumeration. This is trivial to prove in the ZX calculus, and I would like to give such a proof here eventually.  $\square$

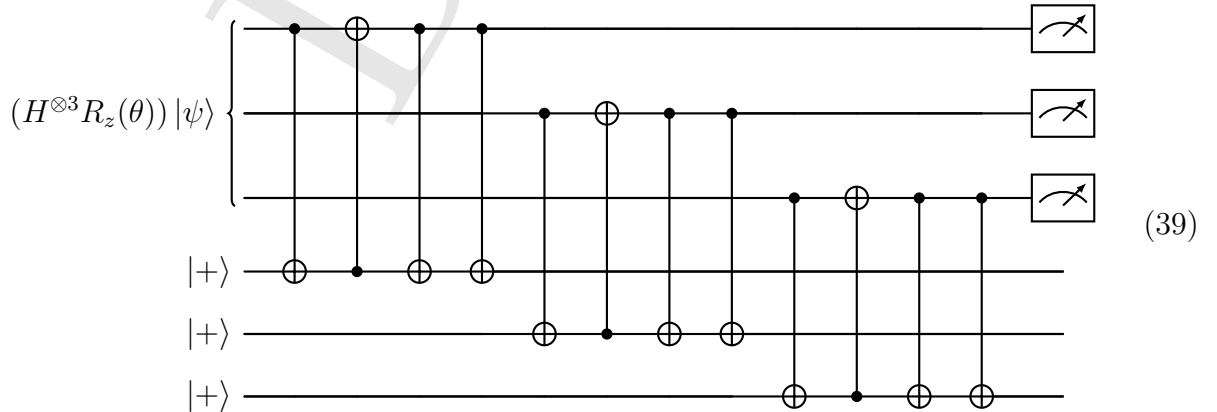
DEFINITION 3.19 We define a gadget that captures the computational power of the construction in Definition 3.17. Start with the standard MBQC protocol:



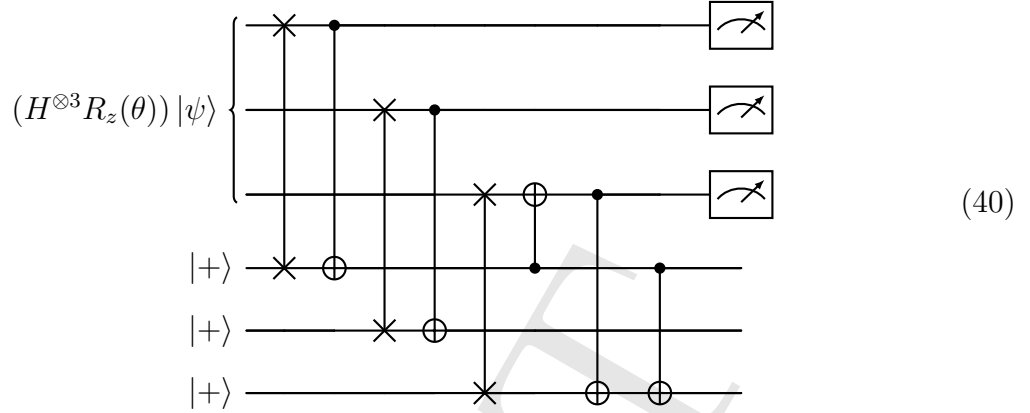
Pushing the gates through like in Equation 14:



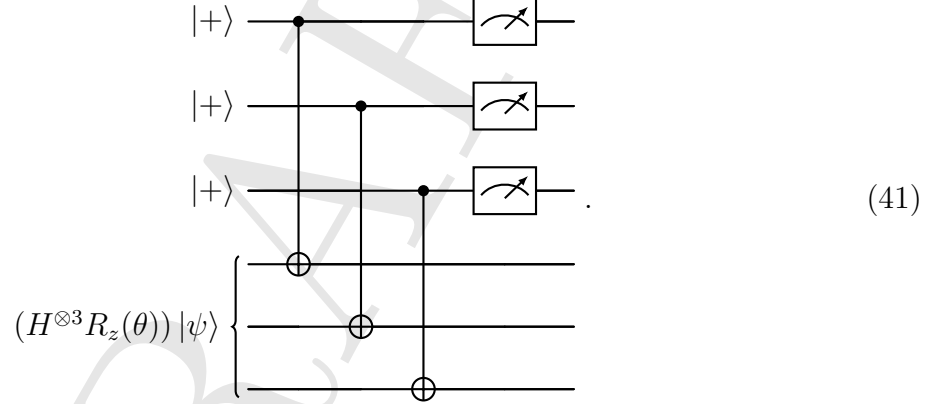
We can add the following CNOTs with affecting the circuit, like in Equation 15:



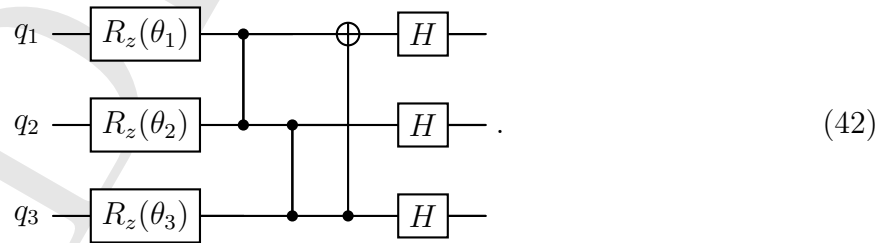
Applying the SWAP gate identity from Equation 16:



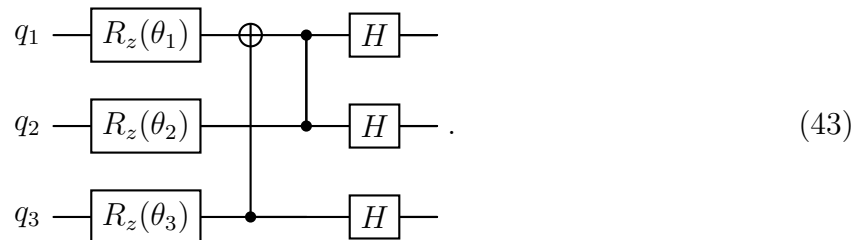
Finally, commuting the swap gates to the beginning gives



Dropping the Pauli gates, we get that we can apply blocks of



Call the above the **width-3 gadget**. Using Fact 3.18, the above gadget simplifies to



### 3.3 Universality of the Width-3 Gadget

DEFINITION 3.20 Without going into details, a **tableau** [11] is an algorithmically efficient way to store and simulate Clifford operations classically.

We can formalize the width-3 gadget in the Clifford tableau by restricting the angles  $\theta_1$ ,  $\theta_2$ , and  $\theta_3$ . This section needs significantly more details and formal definitions. In particular, gate groups and the tableau formalism should be introduced in the Preliminaries section.

FACT 3.21 The group of Clifford operations, denoted  $\langle C \rangle$ , mod the group of Pauli operations, denoted  $\langle P \rangle$ , satisfies

$$|\langle C \rangle / \langle P \rangle| = 1,451,520 \quad (44)$$

FACT 3.22 We repeatedly apply the width-3 gadget with appropriate angles and optional  $S$  gates, representing measurement in the  $Y$  basis instead of the  $X$  basis. Let the group generated by the width-3 gadget and all  $2^3 = 8$  arrangement of  $S$  gates be denoted  $\langle W_3 \rangle$ . Then

$$|\langle W_3 \rangle| = 40,320 = 8!, \quad \frac{|\langle C \rangle / \langle P \rangle|}{|\langle W_3 \rangle|} = 36. \quad (45)$$

*Proof.* By brute force, using Definition 3.20. □

THEOREM 3.23 If the width-3 gadget is universal for quantum computation, this universality must require non-Clifford measurements.

*Proof.* Follows from Fact 3.22. □

We now introduce two new universality tests, one informal and one formal.

DEFINITION 3.24 (Informal universality tests). Replacing the tableau representation in the algorithm behind Fact 3.22 with general matrices, and increasing the number of angles, we should expect that the number of induced gates should increase dramatically. A proof should be given for this.

FACT 3.25 The previous algorithm does not indicate that the width-3 gadget is universal.

THEOREM 3.26 [16] provides an algorithm for determining the universality of a gate set, in terms of a linear algebra problem. Though it is not polynomial in terms of the number of qubits and gates, it is massively parallelizable. Ideally, I would like a GPU kernel that does much of this work.

In conclusion, algorithmic approaches have the potential to uncover subgraph structures using Theorem 3.7. Furthermore, if such gadgets are universal using only Clifford measurements (like the grid), the tableau representation can be used to (somewhat) efficiently prove this through brute force. However, for the width-3 case, it appears that switching to a more theoretical framework may give more results.

## 4 Theoretical Approaches

Theorem 2.40 has potential to prove (or disprove) the universality of CGrid as a resource. However, it will be challenging to find a function  $E$  that fails (satisfies Theorem 2.40) for CGrid yet succeeds for Grid. We formalize this attempt in the next subsection.

### 4.1 LU-LC

DEFINITION 4.1 If  $|G_1\rangle \geq_{\text{LOCC}} |G_2\rangle$  and  $|G_2\rangle \geq_{\text{LOCC}} |G_1\rangle$  then  $E(|G_1\rangle) = E(|G_2\rangle)$  for any entanglement monotone  $E$ . An  $E$  is called **entanglement monotone** if it does not increase, on average, for any LO.

DEFINITION 4.2  $n$ -qubit states  $|\psi\rangle$  and  $|\phi\rangle$  are LU-equivalent if there is a local unitary  $U = \otimes_{i=1}^n U_i$  such that  $|\psi\rangle = U |\phi\rangle$ . We say  $|\psi\rangle =_{\text{LU}} |\phi\rangle$ .

DEFINITION 4.3 The **one-qubit Clifford group** are the 1-qubit unitary matrices generated by  $\langle H, S \rangle$ . They are the stabilizers of the **Pauli group**  $\mathcal{P} = \langle \{-1, 1, -i, i\} \times \{I, X, Y, Z\} \rangle$ .

DEFINITION 4.4  $n$ -qubit states  $|\psi\rangle$  and  $|\phi\rangle$  are LC-equivalent if there is a local  $U = \otimes_{i=1}^n U_i$  such that all  $U_i$  belong to the one-qubit Clifford group and  $|\psi\rangle = U |\phi\rangle$ . We say  $|\psi\rangle =_{\text{LC}} |\phi\rangle$ .

THEOREM 4.5 The LC-equivalence of graph states  $|G_1\rangle$  and  $|G_2\rangle$  is exactly determined by local complementation [cite:need].

DEFINITION 4.6 Let  $G = (V, E)$  be a graph. The local complementation around a vertex  $v \in V$  is done by replacing the induced subgraph (the vertices adjacent to  $v$ , and any edges between these vertices) of  $v$ 's neighborhood with its graph complement.

THEOREM 4.7  $|G_1\rangle =_{\text{LC}} |G_2\rangle$  if and only if there is a sequence of local complementations from  $G_1$  to  $G_2$ . Furthermore, there exists an efficient algorithm to recognize whether graphs are local clifford equivalent [18].

### 4.2 Vertex Minors

DEFINITION 4.8 A graph  $H$  is a vertex minor of a graph  $G$  if there is a sequence of vertex deletions and local complementations from  $G$  to  $H$ .

The above definition is a major motivation for our focus on vertex deletion in the previous section. Thus, it may make sense to swap the order of sections around

THEOREM 4.9 Deciding whether a graph  $H$  is a vertex-minor of  $G$  is NP-complete [5].

DEFINITION 4.10 The definition of the **rank width** of a graph  $G = (V, E)$  is quite involved and given in [13].

FACT 4.11 The rank width of a graph is preserved under local complementation. Vertex deletions may reduce rank-width, so the rank width of  $G$  is at least the rank width of any vertex-minor of  $G$ .

THEOREM 4.12 Every CGrid is a vertex-minor of a graph in every family of graphs with unbounded rank-width [cite:need]. Formally, there is a function  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that every graph of rank-width  $\geq f(n)$  has a vertex-minor isomorphic to  $\text{CGrid}_{n \times n}$  [9].

THEOREM 4.13 Grid graphs contain all graphs as vertex minors. Formally, there is a function  $f : G \rightarrow \mathbb{N}$  such a graph  $G$  is the vertex minor of  $\text{Grid}_{f(G) \times f(G)}$ .

*Proof.* I would like to formalize the proof in Kunal's notes from James George Davies.  $\square$

THEOREM 4.14 If  $\text{CGrid} \geq_{\text{Locc}} \text{Grid}$  then unbounded rank-width is not sufficient for universality.

## 5 Conclusions

We lay out the results of this thesis' algorithms and provide a series of conjectures based on them.

### 5.1 Algorithmic Results

I would like to produce a table of the minimum number of additional edges from attempting to subset  $\text{Grid}_{k \times n}$  from  $\text{CGrid}$ . The cases  $k = 2, 3$  already appear in the previous sections, but  $k = 4, 5$  should be within reach in a few weeks. This is because of the following:

**CONJECTURE 5.1** *This needs to be formalized:* Suppose we have a construction to subset a  $k \times n$  subgraph  $H$  of some  $\text{CGrid}$  with  $a$  additional edges compared to  $\text{Grid}_{k \times n}$ . Suppose further than this construction produces  $b_k$  additional edges for every column added, and this  $b$  remains constant regardless of how many additional columns are added. Then a subgraph  $H'$  of any  $\text{CGrid}$  with additional edges compared to  $\text{Grid}_{k+1 \times n}$  must have at least

$$b_{k+1} \geq \lceil b_k \frac{k+1}{k} \rceil \quad (46)$$

additional edges for every column added.

*Proof.* By contradiction and induction. The base case is  $3 \times n$  where  $b_3 = 1$ , and  $4 \times n$  where if  $b_4 = 1$ , then at least two rows have no additional edge. Deleting the one row connected to this additional edge and performing some transformations afterwards, which need to be formalized leaves a  $3 \times n$  grid with fewer than  $b_3$  additional edges per column. Contradiction. It is not quite clear whether this generalizes and hence is left as a conjecture.  $\square$

**CONJECTURE 5.2** Using the same notation as Conjecture 5.1, whatever construction  $H$  is minimal for  $k$  rows has  $H$  be a bipartite graph. In particular, it is of the form of two antidiagonal regions of the comparability grid. Since the bipartite graphs with  $n$  vertices are counted by  $C_n$ , the  $n$ th Catalan number, which is large but smaller than  $n!$ , significant early termination is still necessary to efficiently find the minimum number of additional edges for any  $k$ .

It is a fact that the number of additional edges must increase as  $k$  increases. In that sense, the  $3 \times n$  case is the easiest case to check universality through algorithmic approaches for.

Here I want to provide a table of the results of the size of the groups generated using the width-3-gadget, and if there is enough compute, the width-4-gadget.

Another algorithmic direction is to include local complementation into the algorithmic search. However this is computationally quite expensive, because it means we cannot use the tuple subset representation of Fact 3.4. Furthermore, (local) graph isomorphism is now necessary to prevent checking the same graph multiple times, which further adds to the complexity of the algorithm.

## 5.2 Future Work

Here I want to provide a table of LOCC operations that can change rank-width. These are meant to be used as a basis to conjecture what to add to local Pauli measurements, local Cliffords, and classical communication.

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